

KUSHAL PITALIYA

Rajkot, Gujarat

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OBJECTIVE

B.Tech ECE graduate with working knowledge of SystemVerilog, UVM-based verification, and RTL design. Looking for a Design Verification role where I can apply my understanding of testbench architecture, protocol debugging, and scripting to contribute to functional verification of digital and mixed-signal IPs.

EDUCATION

Charotar University of Science and Technology (CHARUSAT)

2023 – 2027

B.Tech in Electronics & Communication Engineering

CGPA: 8.74 / 10

Shree G.K. Dholakiya School

Gujarat Board

Higher Secondary (12th) – 62% | JEE: 93 percentile Secondary (10th) – 83%

TECHNICAL SKILLS

Verification: UVM (driver, monitor, sequencer, scoreboard, coverage), Constrained-random verification, Directed testbenches, Assertions (SVA), Functional coverage, Code coverage

HDL / RTL Design: Verilog HDL, SystemVerilog, FSMs (Mealy/Moore), Datapath & Control, Parameterized design

SystemVerilog: `always_ff`, `always_comb`, Interfaces, Clocking blocks, Packages, `rand/constraint`, Enumerated types

Protocols: UART, I2C, SPI, CAN, AMBA (APB/AHB)

FPGA & Simulation: Xilinx Spartan-6, Xilinx ISE / Vivado, ModelSim, Waveform analysis

Scripting & Automation: Python, Shell (Bash), GNU Make, YAML-based automation

Programming: C, C++, Python, Embedded C

Tools: Git / GitHub, Linux (Ubuntu), VS Code, MATLAB

EXPERIENCE

Charotar University of Science and Technology (CHARUSAT)

Aug 2024 – Present

Undergraduate Student Fellow (UGSF) – ECE Department

Anand, Gujarat

- Selected under a merit-based fellowship; contributed to VLSI and embedded systems lab activities.
- Coordinated hands-on workshops on FPGA design, Verilog coding, and PCB prototyping for 50+ students.
- Prepared structured lab documentation and verification procedures for digital design experiments.

KEY PROJECTS

Synchronous FIFO – Design & Verification | *Verilog HDL, SystemVerilog, Spartan-6 FPGA*

Academic

- Designed a parameterized synchronous FIFO with configurable depth and data width for producer-consumer buffering.
- Developed a layered SystemVerilog testbench with separate driver, monitor, and checker components.
- Wrote directed test cases for corner conditions: simultaneous read/write, full-to-empty transition, pointer wraparound.
- Added randomized stimulus to stress-test FIFO behavior under back-to-back burst traffic.
- Verified timing and data integrity through waveform analysis in ModelSim; synthesized and validated on Spartan-6.

FSM Controller – Design & Assertion-Based Verification | *SystemVerilog, SVA*

Academic

- Designed a multi-state FSM (Mealy/Moore) using the 3-block coding style: state register, next-state logic, output logic.
- Wrote SystemVerilog Assertions (SVA) to verify state transition correctness and output timing.
- Tested corner cases: reset during active state, illegal state recovery, and back-to-back transitions.
- Used `always_ff` / `always_comb` strictly to ensure synthesis compatibility.

I2C / SPI Protocol Implementation & Validation | *Embedded C, AVR, Oscilloscope*

Academic

- Implemented bit-bang I2C master and hardware SPI master on AVR; validated bus transactions with logic analyzer.
- Verified protocol timing: setup time, hold time, clock frequency, and ACK/NACK behavior.
- Debugged bus contention and arbitration issues – documented root cause and fix for each failure.

EDA AI Playground – RAG-Based RTL Assistant | *Python, RAG, Prompt Engineering*

Personal

- Built a Retrieval-Augmented Generation engine with curated Verilog/SystemVerilog/UVM knowledge base.
- Designed structured prompts for LLM-based RTL generation: module scaffolding, testbench templates, FSM design.
- Implemented prompt validation pipeline; wrote 24 automated test cases covering knowledge retrieval and prompt assembly.

Ultrasonic Distance Measurement System | *Embedded C, ATmega32, UART, Timers*

Academic

- Designed real-time firmware using interrupt-based architecture: external interrupts for echo capture, Timer1 for pulse-width measurement.
- Validated UART data framing and baud-rate accuracy using oscilloscope; wrote register-level GPIO driver for LCD.

CERTIFICATIONS & ACHIEVEMENTS

Design of Digital Circuits – L&T

iChip 3.0 Verilog Hackathon – Participant (RTL design challenge)

Undergraduate Student Fellow (UGSF) – Merit-based selection, CHARUSAT

Second Runner-up – Idea Show 3.0

Odoo x CHARUSAT Hackathon 2025 – Participant

Workshops: FPGA Design & Implementation, PCB Design & Prototyping